Fiscal Year:	FY 2026	**Task Last Updated:**	FY 05/13/2026
PI Name:	Ivkovic, Vladimir Ph.D.		
Project Title:	Characterizing the Baselines of Sleep Quality, Cognitive / Operational Performance, Immune Function, and Intracranial Fluids for Deep Space Expeditions		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	vivkovic@mgh.harvard.edu	**Fax:**	FY
PI Organization Type:	GOVERNMENT	**Phone:**	617-724-5550
Organization Name:	Massachusetts General Hospital and Harvard Medical School		
PI Address 1:	Department of Psychiatry, Neural Systems Group		
PI Address 2:	149 13th Street, Rm 10.054		
PI Web Page:			
City:	Charlestown	**State:**	MA
Zip Code:	02129	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2017-2018 HERO 80JSC017N0001-BPBA Topics in Biological, Physiological, and Behavioral Adaptations to Spaceflight. Appendix C
Start Date:	04/22/2019	**End Date:**	04/30/2031
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	4	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
Contact Email:	alexandra.m.whitmire@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	FY25 Report: James Ehlers, M.Sc. is no longer with the project.		
COI Name (Institution):	Strangman, Gary E. (Massachusetts General Hospital / Harvard Medical School) Zhang, Quan (Massachusetts General Hospital / Harvard Medical School) Kimberly, William T. (Massachusetts General Hospital / Harvard medical School) Pollonais, JoAnna M.S. (Massachusetts General Hospital / Harvard Medical School) Spielmann, Guillaume (Louisiana State University / School of Kinesiology) Quiriarte, Heather M.Sc. (Louisiana State University / School of Kinesiology) Basner, Mathias M.D., Ph.D. (University of Pennsylvania) Stahn, Alexander (University of Pennsylvania) Crucian, Brian E. Ph.D. (NASA Johnson Space Center)		
Grant/Contract No.:	80NSSC19K0925		
Performance Goal No.:			

Performance Goal Text:

Task Description:

Behavior Associated with Sleep & Immunity in Space (BASIS) [formerly: Characterizing the Baselines of Sleep Quality, Cognitive / Operational Performance, Immune Function, and Intracranial Fluids for Deep Space Expeditions] Sleep is a central physiological regulator of cognitive / behavioral, neurophysiological, and immune functions. The study of sleep quality and duration on orbit may thus yield important insights into etiology and mechanisms of adverse cognitive/behavioral and immunological responses during long duration deep space exploration missions. Deep space and planetary expeditions will expose crewmembers to many risks that could adversely affect mission outcomes. We will use analog-tested cognitive (Cognition) and operational (ROBoT-r) tests for performance assessment and data/sample share for a range of sleep quality and quantity measures, as well as characterizing neuroendocrine and immune functions. By looking across ISS missions of different lengths, we aim to characterize baselines of sleep, cognitive, operational, immune, function. We will collect data on crewmembers participating in Complement of Integrated Protocols for Human Exploration Research (CIPHER), including flyers on short-duration (<3 months), nominal-duration (~6 months) and long-duration (~12 months) missions aboard the International Space Station (ISS). We also plan to study demographically matched control subjects in Earth-based analogs (e.g., the Human Exploration Research Analog, or HERA). Within Complement of Integrated Protocols for Human Exploration Research (CIPHER) project, BASIS is a part of an international project comprising of three synergistic studies lead by Drs. Basner, Stahn, and Ivkovic, respectively. Our specific aims are: (SA1) Characterize cognitive and operational task performance changes as a function of ISS mission duration; (SA2) Characterize cognitive and operational task performance changes as a function of sleep status; (SA3) Characterize cognitive and operational task performance changes as a function of immune activation; (SA4) Characterize cognitive and operational task performance changes as a function of stress; and (SA5) Investigate the interactions between behavior, stress, immunity and sleep. The outcomes of the study will contribute to quantification of crew health and performance risks associated with human spaceflight, and aid in development of technologies for monitoring and mitigating crew health and performance.
BASIS-PNI (Psychoneuroimmunology) Supplement: A supplement to the parent BASIS project, titled Behavior Associated with Sleep & Immunity in Space – Psycho-Neuro-Immunology (BASIS-PNI) was approved by the Human Factors and Behavioral Performance (HFBP) element in June 2020 (around the height of the first COVID wave). The supplement seeks to accelerate the identification and characterization of psychoneuroimmunological interactions that contribute to risks of adverse cognitive/behavioral conditions or mental disorders during exploration class missions. During such missions, crew members will be exposed to numerous stressors including microgravity, prolonged periods of isolation and confinement, environmental extremes, operational demands, and psychosocial challenges. One of the acutely visible and persistent signs of microgravity-induced cephalad fluid shift is facial edema and deformation that consequently reduces an individual's facial expressiveness and capacity to convey non-verbal affective cues. Since facial expression of affective content presents a fundamental aspect of social interaction and maintenance of interpersonal relations, any changes or limitations in this capacity during spaceflight could exacerbate the negative communication or psychological effects of isolation and confinement, and hence contribute to the risk of developing behavioral or psychiatric disorders (BMed). In addition, sustained exposure to psychological stressors, confinement, and isolation are all known to reduce immune competency and promote systemic inflammation and alter facial expressiveness on Earth and in spaceflight. The state of chronic low-grade inflammation is in fact associated with decrements in both physical and psychological functions. Elevation in pro-inflammatory cytokines including IL-1beta, IL-6 and TNF-alpha are associated with increased anxiety and altered mood in healthy adults, as well as reduced processing speed and overall cognition in older adults. It remains unknown whether spaceflight-relevant isolation, confinement, and exposure to operational stressors are associated with inflammation-mediated changes in neurocognitive function and affective processing in the astronaut-like population, nor how these neurocognitive deficits may affect the ability of crewmembers to interpret each-others' altered facial expressiveness due to microgravity-induced facial edema.

BASIS-PNI-BR (Bed Rest) Supplement: A supplement to the parent BASIS project, titled Behavior Associated with Sleep & Immunity in Space – PsychoNeuroImmunology in Head Down Tilt Bedrest (BASIS-PNI-BR) was approved by the Human Factors and Behavioral Performance (HFBP) element in April 2024. The primary goal of this supplement is to characterize the relationship between perceived emotional valence based on previously recorded facial expression photographs—collected head up/head down tilt— and immune function before, during, and after exposure to 60-day head down tilt (HDT) bedrest (BR)—conducted at DLR :envihab facility. This data from HDTBR analog would be synergistic to the data collected in the C7 HERA campaign which is a part of the PNI supplement.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

BASIS: Successful completion of this project will be a milestone in spaceflight behavioral, neurophysiological, and immune investigation. It will be the largest study of operational performance in space, as well as the first to conduct in-flight resting-state and task-related functional brain imaging. In addition, it will be the first study to probe the relationships and interactions between behavioral, neurophysiological, sleep, and immune functioning. The use of three different length missions will help us to identify effects or trends that may extrapolate towards longer missions, with the known caveats of such extrapolation. The findings will be important on Earth as well, by helping to better understand the complex inter-relationships between sleep, brain physiology, immune function, and cognitive performance.
BASIS-PNI: Findings from the PNI study will directly contribute to identification of key threats to behavioral health (BMed3), development of measures for monitoring behavioral health (BMed2), and understanding interactions between immune, psychological and perceptual factors in behavioral health associated with long duration missions.

BASIS-PNI-BR: Findings from this study will further characterize the effects of HDT bedrest on neurobehavioral and immunological function, and thus directly contribute to identification of key threats to behavioral health, development of measures for monitoring behavioral health, and understanding interactions between immune, psychological and perceptual factors in behavioral health associated with long duration missions. By using the SAFER survey – developed as a part of the BASIS-PNI project and tested in HERA, this study will further validate this measure in another spaceflight analog – HDT BR – thus expanding available tools for assessment of health and human performance in spaceflight.

Task Progress:	During the 7th year of this project, we focused on the following major tasks: • Continuation of integration, recruitment, and data collection on CIPHER-BASIS. • Completing preliminary analyses of CIPHER-BASIS data on operational performance (ROBoT-r); During the report period, we collect data in pre-flight, and in-flight phases of International Space Station (ISS) missions. This included collection of ROBoT-r operational performance data, as well as Cognition, Spatial Cognition, Standard Measures sleep questionnaires, and actigraphy. In addition, stress and immune biomarkers were being collected from salivary and serum samplings. These included PBMC cytokines (IL-6, TNF-alpha, IFN-gamma, IL-10, IL-12 p70, 13-plex cytokine); Cortisol, DHEA; as well as plasma and salivary analytes via data sharing between Coronary Responses (PI: Dr. Levine) and Vascular Calcium (PI: Dr. Hughson) CIPHER projects. Data is being processed at the levels of the entire CIPHER study and will become available in the coming weeks or months. Preliminary data on ROBoT-r operational performance has been presented at 2026 NASA Human Research Program Investigators' Workshop (HRP IWS). • Completed preliminary data analyses from HERA C7 (M1-M4) within BASIS-PNI; After completing Specific Aims 1 and 2 in the previous reporting periods, during the current reporting period we have continued to work on finalizing Specific Aim 3. During the reporting period, we completed preliminary analyses of behavioral data from the BASIS-PNI study in HERA C7. This included work on analyzing available behavioral data, and preparation of biological samples for analyses in collaboration with Dr. Spielmann's and Dr. Crucian's teams. Behavioral and biological data was previously collected during Missions 1-4, on a sample of N=16 HERA crewmembers, 50% female, 37±5.5 years of age. All data are being collected before (MD-1), during (MD23, MD37), and after (MD+1) missions within HERA C7, and included: Behavioral: Emotional state (Positive Affect Negative Affect Scale; PANAS), anxiety (State Trait Anxiety Index (STAI), Spaceflight Analog Facial Recognition (SAFER), Pittsburgh Sleep Quality Index (PSQI), and Cognition battery; Biological: Resting saliva sample to measure stress responses cortisol, DHEA-S, epinephrine, norepinephrine, dopamine, alpha amylase, lysozyme, and chromogranin A (ELISA); Resting serum sample to measure immune and stress response--circulating cytokines (22-plex). Preliminary analyses were completed on behavioral data, while analyses of biological data are ongoing. Full psychoneuroimmunological analyses will be performed next, pending completion of biological data analyses. • Completed data collection and preliminary data analyses in Campaigns 1 and 2, and continued preparation for Campaign 3 of the BASIS-PNI-BR supplement project; Over the course of the reporting period, we have continued to collaborate and coordinate with NASA SMC ROI and NASA/DLR/:envihab teams (Stefan Moestl, Sarah Piechowski, Edwin Mulder, Bryan Caldwell, etc.) to complete research integration, data collection and data sharing in NASA/DLR/:envihab SMC Campaigns 1 and 2. In addition we have been working with support from the Behavioral Health and Performance (BHP) laboratory at NASA Johnson Space Center (Diana Arias, Sheen Dev, Princess Nafiyah, Suzanne Bell), as well as HHPIT team (Ahmed El-rasheedy, Syed Arif Rahman, Cynthia Fontenot). These activities also included continuously checking hardware and software during deployment to DLR/:envihab, and conducting continuous quality controls on the incoming data. We are currently executing a data sharing agreement which would allow transfer of data from DLR/:envihab that includes outcomes on Positive Affect Negative Affect Scale (PANAS) and the General Health Questionnaire (GHQ). Biomarkers of immune and stress function are scheduled to be delivered at the end of the data collection period, and cannot be included in analyses at this time. This included collection of neurobehavioral and general health data, particularly on emotional state (Positive Affect Negative Affect Scale; PANAS), anxiety (State Trait Anxiety Index (STAI), General Health Questionnaire (GHQ), Pittsburgh Sleep Quality Index (PSQI), Spaceflight Analog Facial Emotion Recognition (SAFER), as well as demographic data. In addition, stress and immune biomarkers were being collected via salivary and serum samplings. Plasma samples from all the subjects will also be analyzed in duplicate for cytokines and biomarkers of inflammation and stress including cortisol, DHEA-S, epinephrine, norepinephrine, dopamine, alpha amylase, lysozyme, and chromogranin (NULISA-seq). All saliva samples will be collected using the non-invasive Salimetrics polyester Oral Swab (SOS) technology (Salimetrics, CA, USA) which has been validated for the measurement of salivary hormones, sIgA and alpha-amylase. We used the same statistical approaches described in BASIS-PNI study description. • Educational activities for students on BASIS-PNI, and BASIS-PNI-BR projects; BASIS-PNI: We trained 2 graduate student interns on the BASIS-PNI supplement project. The students participated in literature review and preparation of review manuscript focusing on psychoneuroimmunological aspects of neurobehavioral changes in spaceflight. The manuscript is in preparation. The students participated in interpretation of processed data, and preparation of manuscripts as well as conference presentations. The students were included as co-authors and participated in development and delivery of the BASIS-PNI poster presentation at HRP IWS 2026. One graduate student completed their degree during the report period. | BASIS-PNI-BR Project: Two graduate students contributed to the preparation of BASIS-PNI-BR supplement project. The students participated in interpretation of processed data, and preparation of manuscripts as well as conference presentations. The students were included as co-authors and participated in the development and delivery of the BASIS-PNI-BR poster presentation at HRP IWS 2026. • Presented 2 talks (CIPHER-BASIS, CIPHER-BASIS-PNI-BR) and 1 poster (CIPHER-BASIS-PNI) at the 2026 HRP-IWS, contributed to talks by Dr. Basner and Dr. Stahn who we collaborate within the Spatial Cognition CIPHER project.
Bibliography Type:	Description: (Last Updated: 05/13/2026)
Abstracts for Journals and Proceedings	Zhang Q, Ivkovic V, Spielmann G, Vujovic N, Kimberly WT, Strangman GE. "Characterizing the baselines of sleep quality, cognitive/operational performance, immune function, and intracranial fluids for deep space expeditions." 2020 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 27-30, 2020. Abstracts. 2020 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 27-30, 2020. , Jan-2020
Abstracts for Journals and Proceedings	Ivkovic V, Zhang Q, Spielmann G, Skokin K, Manoach M, Tomy N, Strangman. "Preliminary investigations of psychoneuroimmunological interactions that contribute to risks of adverse cognitive/behavioral conditions in spaceflight." 2021 NASA Human Research Program Investigators' Workshop, Virtual, February 1-4, 2021. Abstracts. 2021 NASA Human Research Program Investigators' Workshop, Virtual, February 1-4, 2021. , Feb-2021

Abstracts for Journals and Proceedings	Zhang Q, Ivkovic V, Spielmann G, Vujovic N, Kimberly WT, Strangman GE. "Characterizing the baselines of sleep quality, cognitive/operational performance, immune function, and intracranial fluids for deep space expeditions." 2021 NASA Human Research Program Investigators' Workshop, Virtual, February 1-4, 2021. Abstracts. 2021 NASA Human Research Program Investigators' Workshop, Virtual, February 1-4, 2021. , Feb-2021
Abstracts for Journals and Proceedings	Ivkovic V, Peters D, Schenkel M, White B, Cumani J, Thoolen S, Zhang Q, Spielmann G, Strangman G. "CIPHER-NINSCAN-PNI: Quantifying facial deformation in HUT/HDT for assessing psychoneuroimmunological contributions to risks of adverse cognitive/behavioral conditions in spaceflight." 2022 NASA Human Research Program Investigators' Workshop, Virtual, February 7-10, 2022. Abstracts. 2022 NASA Human Research Program Investigators' Workshop, Virtual, February 7-10, 2022. , Feb-2022
Abstracts for Journals and Proceedings	Zhang Q, Strangman G, Ivkovic V, Spielmann G, Kimberly W, Vujovic N. "NINscan: Characterizing the baselines of sleep quality, cognitive/operational performance, immune function, and intracranial fluids for deep space expeditions." 2022 NASA Human Research Program Investigators' Workshop, Virtual, February 7-10, 2022. Abstracts. 2022 NASA Human Research Program Investigators' Workshop, Virtual, February 7-10, 2022. , Feb-2022
Abstracts for Journals and Proceedings	Ivkovic V, Basner M, Stahn AC, Spielmann G, Pollonais J, Vujovic N, Zhang Q, Strangman GE. "Temporal nature of cognitive and visuospatial brain domain changes during long-duration low-Earth orbit missions: Effects of sleep quality, immune function, and stress on operational performance (Project C)." 2023 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 7-9, 2023. Abstracts. 2023 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 7-9, 2023. , Feb-2023
Abstracts for Journals and Proceedings	Ivkovic V, Peters D, Cumani J, Pollonais J, Spielmann G, Thoolen S, White BM, Zhang Q, Strangman GE. "CIPHER-NINscan-PNI: Quantifying facial deformation in HUT/HDT for assessing psychoneuroimmunological contribution to risks of adverse cognitive/behavioral conditions in spaceflight – 2023 update." 2023 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 7-9, 2023. Abstracts. 2023 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 7-9, 2023. , Feb-2023
Abstracts for Journals and Proceedings	Stahn AC, Basner M, Ivkovic V, Barsby P, Besnard S, Brauns K, Denise D, Dinges DF, Friedl-Werner A, Gunga HC, Garthe A, Hartley T, Hetzer S, Miller N, Riecke BE, Spielmann G, Stangl M, Strangman GE, Wolbers T, Kühn S. "CIPHER spatial cognition: Visuo-spatial abilities and their neural basis during long-duration low-Earth orbit missions." 2023 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 7-9, 2023. Abstracts. 2023 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 7-9, 2023. , Feb-2023
Abstracts for Journals and Proceedings	Basner M, Stahn A, Bell S, Ivkovic V, Dinges DF, Ecker AJ, Gilad Y, Moore TM, Roalf D, Gur RC. "Temporal nature of cognitive and visuospatial brain domain changes during long-duration low-Earth orbit missions: Neurostructural and cognitive changes (Project A)." 2023 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 7-9, 2023. Abstracts. 2023 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 7-9, 2023. , Feb-2023
Abstracts for Journals and Proceedings	Ivkovic V, Basner M, Stahn A, Spielmann G, Pollonais J, Zhang Q, Strangman G. "Temporal nature of cognitive and visuospatial brain domain changes during long-duration low-Earth orbit missions: Baselines of sleep, cognitive/operational performance, and immune function (Project C)." 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. , Feb-2024
Abstracts for Journals and Proceedings	Ivkovic V, Ehlers J, Pollonais J, Spielmann G, Thoolen S, White B, Zhang Q, Strangman G. "CIPHER-PNI: Quantifying facial deformation in HUT/HDT for assessing psychoneuroimmunological contribution to risk of adverse cognitive/behavioral conditions in spaceflight - 2024 update." 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. , Feb-2024
Abstracts for Journals and Proceedings	Basner M, Bell S, Dinges D, Ecker A, Gilad Y, Gur R, Ivkovic V, Jones C, Moore T, Roalf D, Ruparel K, Stahn A. "Temporal nature of cognitive and visuospatial brain domain changes during long-duration low-Earth orbit missions: Neurostructural and cognitive changes (Project A)." 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. , Feb-2024
Abstracts for Journals and Proceedings	Stahn A, Basner M, Ivkovic V, Barsby P, Besnard S, Brauns K, Denise P; Dinges D, Friedl-Werner A, Gunga H-C, Hartley T, Hetzer S, Garthe A, Miller N, Riecke B, Spielmann G, Stangl M, Strangman S, Wolbers T, Kühn S. "CIPHER spatial cognition: Visuo-spatial abilities and their neural basis during long-duration low-Earth orbit missions." 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. , Feb-2024
Abstracts for Journals and Proceedings	Ivkovic V, Basner M, Stahn A, Spielmann G, Quiriarte H, Pollonais J, Zhang, Strangman GE. "Temporal Nature of Cognitive and Visuospatial Brain Domain Changes during Long Duration Low Earth Orbit Missions: Baselines of Sleep, Cognitive/Operational Performance, and Immune Function (Project C)." 2026 NASA Human Research Program Investigators' Workshop, Virtual, April 7-9, 2026. Abstracts. 2026 NASA Human Research Program Investigators' Workshop, Virtual, April 7-9, 2026. , Apr-2026
Abstracts for Journals and Proceedings	Basner M, Stahn A, Jones C, Bell S, Ivkovic V, Ecker A, Ruparel K, Moore T, Roalf D, Gur R. "Temporal nature of cognitive and visuospatial brain domain changes during long-duration low-Earth orbit missions." 2026 NASA Human Research Program Investigators' Workshop, Virtual, April 7-9, 2026. Abstracts. 2026 NASA Human Research Program Investigators' Workshop, Virtual, April 7-9, 2026. , Apr-2026

Abstracts for Journals and Proceedings	Ivkovic V, Pollonais J, Spielmann G, Quiriarte H, Bostian C, Piechowski S, Moestl S, Mulder E, Grandchamps G, Morana R, Kanka MH, White B, Zhang Q, Strangman GE. "Behavior associated with sleep and immunity in space: Psychoneuroimmunology in head-down tilt bedrest (preliminary data from campaigns 1 and 2)." 2026 NASA Human Research Program Investigators' Workshop, Virtual, April 7-9, 2026. Abstracts. 2026 NASA Human Research Program Investigators' Workshop, Virtual, April 7-9, 2026. , Apr-2026
Abstracts for Journals and Proceedings	Ivkovic V, Spielmann G, Quiriarte H, Crucian B, Morana R, Bostian C, Grandchamps G, Kanka MH, White B, Zhang Q, Strangman G. "Behavior associated with sleep and immunity in space: Psychoneuroimmunology in HERA campaign 7." 2026 NASA Human Research Program Investigators' Workshop, Virtual, April 7-9, 2026. Abstracts. 2026 NASA Human Research Program Investigators' Workshop, Virtual, April 7-9, 2026. , Apr-2026
Abstracts for Journals and Proceedings	Stahn A, Barsby P, Ivkovic V, Basner M. "Impairments in visuo-spatial performance in response to spaceflight: First insights from CIPHER spatial cognition." 2026 NASA Human Research Program Investigators' Workshop, Virtual, April 7-9, 2026. Abstracts. 2026 NASA Human Research Program Investigators' Workshop, Virtual, April 7-9, 2026. , Apr-2026